

Experimental Report

Distance recovery, orientation recovery, and scaffold characterisation
Runs of 13 August 2026

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Abstract

Three experiments, run and reported against predictions stated beforehand. Together they establish the two links that carry the scientific claim of `boundary_recovery_v5` — that the *distance* to a hidden routing boundary and its *direction* are both recoverable from query-only observations — and they characterise the real adversarial target the method is aimed at.

Of seven predictions, four were confirmed and three falsified. The falsifications were the more useful half: one inverted a design axis of the next experiment, one moved the mixture likelihood-ratio test onto the critical path, and one retired the explicit falsification condition that could have sunk the whole approach.

Provenance tags.

[VALIDATED]	measured against ground truth
[SPECIFIED]	designed, not yet run
[NEW]	new finding from these runs
[OPEN]	acknowledged gap
[LIMIT]	structural impossibility

1 Summary

What the three experiments jointly establish. From query access alone, at an anchor near a hidden additive gate, both the *distance* to the boundary and its *direction* are recoverable with quantified error; per-anchor direction error is pure variance, so it pools away; and against the published adversarial target the probes are neither recognised by its detector nor rejected by our own on-manifold filter.

PART I

Experiment 1 — the distance estimator

#	Prediction, as stated	Outcome
E1-a	$\hat{d} = -\sigma\Phi^{-1}(\hat{\pi})$ recovers true distance	CONFIRMED. $r = 0.9958$, median relative error -0.92% .
E1-b	implied distance is constant across rungs for a gate and wanders for a smooth confound	CONFIRMED. CV 0.098 against 0.56–1.24.
E2-a	misassignment attenuates but does not rotate: bias ≈ 0 , variance up	CONFIRMED. Bias not significant in 0 of 20 cells.
E2-b	orientation error is U-shaped in π , optimum near 0.15–0.30	FALSIFIED. Monotone <i>increasing</i> ; low π is best.
S-1	the OOD detector calls $> 80\%$ of our probes real	CONFIRMED on 3 of 4 (0.98 / 0.91 / 0.83 against 0.000 for LIME). Missed on Communities & Crime.
S-2	the scaffold’s $\Delta/\sigma_{\text{resid}}$ exceeds 2.5	FALSIFIED. 1.60 / 1.88 / 1.95 — at the dip’s floor.
S-3	retention falls below ρ_{min} ; the audit abstains on its own target	FALSIFIED. $\rho = 0.985\text{--}0.997$, zero anchors below threshold.

2 Question

A two-component fit at an anchor returns a mixing fraction $\hat{\pi}$: the share of probes that landed on the far side of the gate. For a probe cloud $z \sim N(0, \sigma^2 I)$ and a unit normal n , the signed projection $z^\top n$ is $N(0, \sigma^2)$, so crossing at distance d occurs with probability $\Phi(-d/\sigma)$. Read backwards,

$$\hat{d}_s = -\sigma_s \Phi^{-1}(\hat{\pi}_s)$$

The crossing law itself is validated to three decimals (E1a, July). Using it as an *estimator* of distance to an unobservable boundary had never been tried.

3 Design

The registered 1-D known-regimes setting: gate at $x = 0.5$ with $\Delta = 0.30$, honest slope 0.15, ladder `geomspace(0.02, 0.2, 3)`, $m = 1000$ probes per rung. The stored `sim1d_anchors.csv` keeps only a *median* of $\hat{\pi}$ across scales, so the inversion was not testable from it; the experiment reruns the protocol recording $\hat{\pi}$ per scale.

Orientation is query-only: the anchor’s own response says which mixture component it sits in, and the crossing mass is the other one. Six models — gated, honest, a kink control, and three Gaussian-process nulls with lengthscales above and below the ladder top — across a $10\times$ noise sweep, 3,218 anchor-by-rung rows.

Two guards, both computable without ground truth: the dip fires under Bonferroni, and $\hat{\pi} \in [0.05, 0.50]$.

4 Results

$\text{corr}(\hat{d}, d_{\text{true}}) = \mathbf{0.9958}$ ($n = 217$) · median relative error $\mathbf{-0.92\%}$ · median absolute error 0.0031

Tracks correctly across every bin from $d = 0.012$ to 0.251, and is flat across a $10\times$ noise

sweep: median absolute error 0.0029/0.0031/0.0027 at $\tau = 0.005/0.02/0.05$.

Relative error degrades close to the boundary — 3.3% beyond $d = 0.1$, 5.0% at 0.05–0.1, 26% below 0.01 — where the *absolute* error is still ~ 0.003 and only the denominator is small. Report absolute error near the boundary.

The second invariance. [NEW] For a planar boundary at fixed distance, $\hat{d}_s$ should not depend on the rung. It does not for a gate, and does for a smooth confound:

Model	median CV of $\hat{d}$	fire rate
gated	0.098 (0.074 at $\tau = 0.02$)	0.678
GP null, $\ell = 0.05$, $A = 0.10$	0.557	0.397
GP null, $\ell = 0.05$, $A = 0.20$	1.042	0.114
kink control	1.241	0.023
honest	—	0.000
GP null, $\ell = 0.30$ (above ladder top)	—	0.000

At $CV < 0.15$: 56% of gates retained, 93% of confounders rejected. As a standalone filter that is expensive; its value is as one of three orthogonal checks, and the operating point should be set once the full pipeline exists.

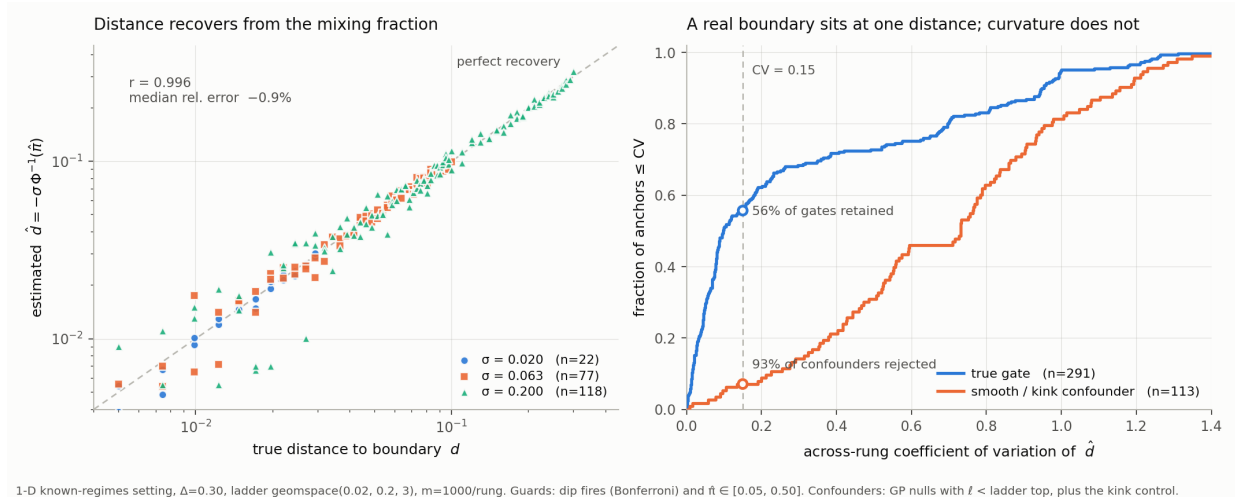


Figure 1: Left: recovered against true distance, by probe scale. Right: across-rung constancy of $\hat{d}$ separates a gate from smooth and kink confounders.

5 What this did not establish

The estimate is **conditional on detection**. Without the dip guard, anchors beyond the ladder top return confident nonsense — true 0.277 against estimated 0.141, with enormous spread — because $\hat{\pi}$ there is not a crossing fraction at all. The guard is query-only, so the conditioning is legitimate, but it belongs in the claim.

And this ran the **naïve** protocol: 1-D, ambient Gaussian probes, no tangent frame, no trimming, no density filter. It validates the inversion, not the inversion inside the pipeline. [OPEN]

PART II

Experiment 2 — the orientation estimator

6 Question

The Stage-3 separability guard fits a linear classifier on (probe coordinates, mixture responsibilities). Its decision boundary, lifted through the tangent frame, is an estimate of the gate’s normal — so test and recovery are the same computation. The concern this had to answer: **responsibilities are estimated, not true branch labels**, so the chain

$$\text{mixture misassignment} \rightarrow \text{classifier label noise} \rightarrow \text{bias in } \hat{n}$$

could dominate everything else near weak separation.

The right figure of merit is not mean error but the **bias/variance split**. Independent unbiased errors shrink as $1/\sqrt{N}$ under pooling — 40° per anchor becomes $\sim 4^\circ$ at $N = 100$. A systematic 5° rotation is still 5° after 200 anchors, and propagates straight into $\hat{t} = \hat{c}/\hat{n}_i$.

7 Design

Intrinsic $d = 2$ in ambient $D = 20$, flat manifold, planted hyperplane, frame supplied (Stage 0 deferred). The true normal is drawn per anchor so no coordinate is privileged; error is $\angle(\hat{n}, n)$ modulo sign. 200 anchors per cell over a $\Delta/\tau \times \pi$ grid, $m = 1000$ probes per anchor, 4,000 anchors total.

The instrument is an oracle contrast: fit the discriminant twice at each anchor — once on estimated responsibilities, once on the *true* branch labels, which exist by construction — and difference the orientation errors. The oracle arm is the variance floor; the gap is the misassignment cost.

8 Results

Misassignment costs variance, not bias. Mean signed rotation per cell: the oracle arm lies in $[-0.35^\circ, +0.27^\circ]$; the estimated-responsibility arm lies in $[-6.6^\circ, +5.9^\circ]$ and is **not significant at 95% in any of the 20 cells**.

Pooled orientation error falls as $1/\sqrt{N}$: at $\pi = 0.05$, from 6.8° to 0.9° between $N = 1$ and $N = 100$ at $\Delta/\tau = 2.5$, and $20.4^\circ \rightarrow 1.8^\circ$ at $\Delta/\tau = 1.5$ — below the dip’s floor entirely.

Why there is no rotation. The result has a mechanism rather than being empirical luck. For $z \sim N(0, \sigma^2 I)$ and a half-space boundary, both class-conditional means of z lie *exactly* along n : the components orthogonal to n are independent of $z^\top n$, so their conditional means vanish. Since the responsibilities are a function of the one-dimensional residual, and the residual is a function of

$z^\top n$, the responsibilities are a function of $z^\top n$ alone — so corrupted labels attenuate the separation without turning it.

Prediction E2-b falsified, and in the opposite direction. Orientation error is **monotone increasing** in π , not U-shaped. Per-anchor standard deviation at $\Delta/\tau = 5$ runs 3.9° at $\pi = 0.05$ to 47.5° at $\pi = 0.50$. The reasoning behind the prediction — few crossers means a noisy discriminant — was wrong because the crossers sit further out along the normal, and the separation of class-conditional means *rises* as π falls:

π	0.05	0.10	0.20	0.35	0.50
mean separation along n	2.17σ	1.95σ	1.75σ	1.63σ	1.60σ

At $\pi \rightarrow 0.5$ there is additionally no majority branch for LTS to trim toward.

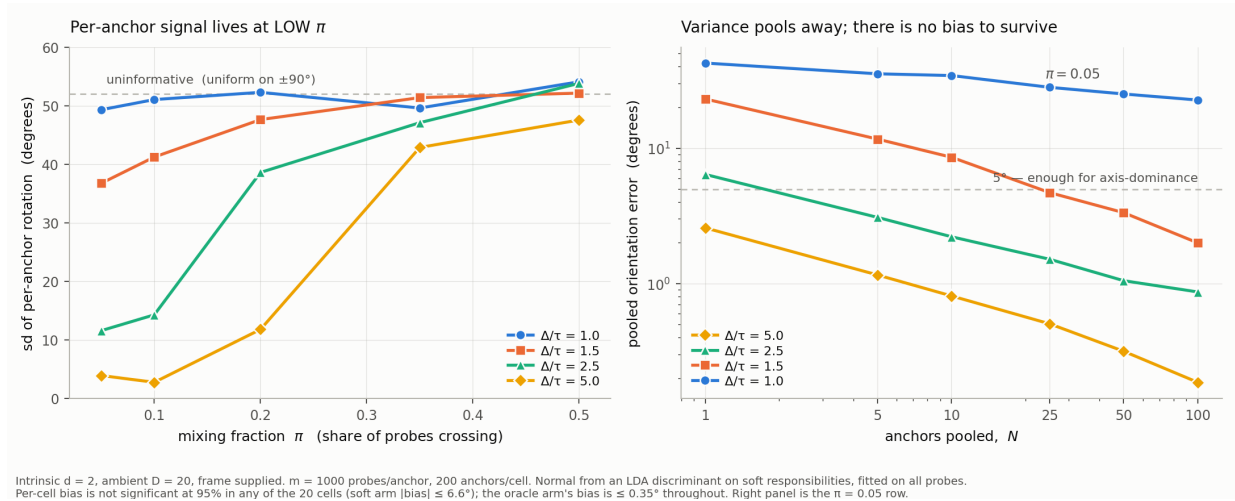


Figure 2: Left: per-anchor information against the mixing fraction; 52° is what uniformly random directions give. **Right:** pooled error against anchors, at $\pi = 0.05$.

9 Consequences for the method

[NEW]

- **Fit the discriminant on all probes, not the LTS inlier set** — 5.8° against 21.9° median error. The trim removes points that are predominantly the minority branch, which is exactly the branch carrying the crossing signal. Trim for the *mixture*; discard the mask for the *discriminant*.
- **Soft responsibilities beat hard $\gamma > 0.5$ labels** — 5.8° against 7.6° .
- **Useful anchors sit at low π , i.e. $d/\sigma \in [1.28, 1.64]$.** This inverts Experiment P's distance axis, whose small values placed anchors close to the boundary. It also agrees with the distance estimator, so one rung serves both.

A no-information guard is now required. Outside the operating region ($\Delta/\tau = 1.0$ at any π , or $\pi \gtrsim 0.35$) the per-anchor rotation has standard deviation $\approx 52^\circ$ — what uniformly

random directions give. Pooling uninformative *axial* estimates does not converge to the truth and does not converge to nothing: the mean direction of N random axes is itself uniformly random, so the pooled estimate lands on an arbitrary direction with an interval that *tightens* around it. Pooling can be confidently wrong in a way per-anchor abstention never is.

10 What this did not establish

Intrinsic $d = 2$ only. The discriminant is a direction in $\mathbb{R}^{\hat{d}}$, so error should scale with *intrinsic* dimension; E1’s dimension-independence result concerns *ambient* D and does not transfer. [OPEN]

The pooled curve was simulated by averaging rotations measured *relative to known truth*. Real pooling has no reference and must find the mode; in $d = 2$ with a common normal the variance behaviour transfers, but the estimator itself and the offset $\hat{c}$ remain untested. That is Experiment P.

PART III

Experiment 3 — scaffold characterisation

11 Question

Every power figure in the project is measured against gates we planted ourselves. The target is the Slack et al. (AIES 2020) scaffolding attack: a biased f , an innocuous ψ , and a random forest trained to separate real data from LIME-perturbed samples, routing between them.

Two measurements decide what the rest of the work is for. **S5**: what fraction of the audit’s own probes does the detector call real? LIME perturbs every feature ambiently with $N(0, 0.3)$; ours is isotropic inside a $\hat{d}$ -dimensional tangent frame at the local geometric scale — a distribution the detector was never trained against. **S1**: what is the scaffold’s actual $\Delta/\sigma_{\text{resid}}$?

12 Design

Four targets: our geospatial housing scaffold, and COMPAS, German Credit and Communities & Crime through Slack’s own `adversarial_models`. 150 anchors per target, 400 probes per anchor, frame and radius supplied by local PCA on the 60 nearest real points. Probes are generated in standardised space and mapped back to raw units, because raw units are what the detector sees. A radius sweep over $\{0.02 \dots 1.00\} \times$ the local k -NN scale turns S5 from a yes/no into a curve.

13 Results

S4 (detector leakage on real points) is 0.000 everywhere, and **no anchor on any target** had ρ below $\rho_{\min}$.

Target	S5 audit	S5 LIME	S3 ρ	straddle rate	response values
COMPAS	0.980	0.000	0.997	0.013	1
German	0.900	0.000	0.987	0.253	2
Housing	0.860	0.000	0.996	0.473	400
Communities & Crime	0.001	0.000	0.985	0.007	2

The detector identifies 100% of LIME’s own perturbations and misses 83–98% of ours. It was trained to recognise exactly one perturbation family and generalises to no other.

Housing resolves to trichotomy regime (ii). Across the radius sweep, invisibility decays gently ($0.94 \rightarrow 0.83$) while the fraction of anchors whose ball straddles the OOD boundary rises sharply ($0.10 \rightarrow 0.53$). Both conditions hold at once: at full k -NN radius, 83% of probes are invisible and about half of anchors reach the gate. COMPAS is regime (i) — invisible but never crossing. German is mixed, at 25%.

S-2 falsified, and it costs. $\Delta/\sigma_{\text{resid}}$ is 1.60 (COMPAS), 1.88 (German), 1.95 (housing) — not the predicted >2.5 . The real scaffold sits *at* the dip’s floor, where measured power is 0.000, and inside the equal-variance LRT’s 0.42–0.54 band. **The LRT is therefore no longer optional, and neither is pooling.** Its threshold has never been calibrated through the pipeline; that moves onto the critical path.

S-3 falsified, and it is the good kind. The explicit falsification condition — that the density filter abstains exactly where the signal lives — did *not* fire. Retention is 0.985–0.997 on every target with zero anchors below threshold. This was the single result that could have invalidated the on-manifold approach against this target.

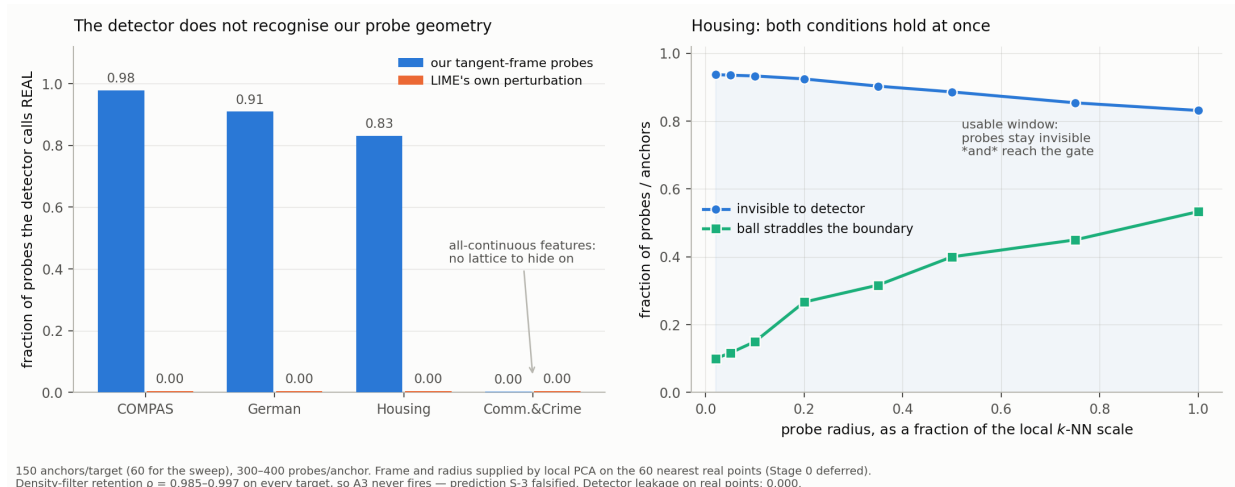


Figure 3: Left: probe visibility, ours against LIME’s own. **Right:** the housing operating window — invisibility and boundary-crossing hold simultaneously.

14 The Communities & Crime miss

Not a radius or frame artifact: visibility stays at 0.011 with the dimension cap raised to 30, and at 2% of the k -NN radius. The mechanism appears to be a **lattice** effect. Adding 10^{-3} of noise to a training point drops “called real” to 0.000/0.000/0.046, while genuinely held-out real points are still called real at ≈ 1.0 — so the detector is not memorising; it keys on off-grid movement in quantised coordinates.

Tangent-frame probing largely respects that grid, because low-dimensional local PCA on mixed tabular data concentrates on the continuous directions. Off-lattice fraction: COMPAS 0.33, housing 0.62, German 0.69, **Communities & Crime** 0.96 — and its features are nearly all continuous, so there is no grid to stay on.

[OPEN] The ordering is right but this is not a clean single-factor account: German is 0.69 off-lattice and still reaches 0.91 visibility, so ambient dimension (101 against ≤ 29) is plausibly also at work. Directionally supported, not established.

15 Two consequences

A discrete-response Stage 3 is a confirmed prerequisite for the Slack datasets. They expose **one or two** distinct response values; housing gives 400. Stage 3 tests the shape of a continuous residual distribution and does not apply to a binary response. Anticipated in the 31 July review; now measured.

A sharper claim is available. Scaffolding attacks keyed on LIME’s ambient Gaussian perturbation are defeated by any probe family respecting the data’s discrete structure — *but not on all-continuous data*. That is a statement about the attack’s scope, and Communities & Crime is the counterexample that makes it non-trivial.

16 What this did not establish

Experiment S showed the gate is **reachable**. It detected and recovered *nothing*. The end-to-end run against the housing scaffold is a separate piece of work and is now the clearest route to the second deliverable.

PART IV

Consolidated

17 What changed in the method

1. The distance estimator $\hat{d} = -\sigma\Phi^{-1}(\hat{\pi})$ enters Stage 4. [NEW]
2. Across-rung constancy of $\hat{d}$ enters Stage 5 as a second invariance, orthogonal to the $\hat{\Delta}$ scaling exponent. [NEW]
3. The discriminant fits on all probes, not the LTS inlier set. [NEW]
4. Anchor placement targets $\pi \leq 0.10$; Experiment P’s distance axis inverts. [NEW]
5. A no-information guard becomes part of Experiment P. [NEW]

6. LRT calibration through the pipeline moves onto the critical path, because the real target sits at the dip's floor. [\[NEW\]](#)
7. The density-filter falsification condition is retired. [\[NEW\]](#)

18 Reproduction

All seeded; all under `sim/` in `detect-recover-interpret`.

Experiment	Script	Outputs
1 — distance	<code>exp_distance_estimator.py</code>	<code>distance_estimator_rows.csv</code> (3,218)
2 — orientation	<code>exp_normal_estimator.py</code>	<code>normal_estimator_rows.csv</code> (4,000); <code>normal_estimator_</code> <code>probelog.csv.gz</code>
3 — scaffold	<code>exp_s_scaffold.py</code> <code>exp_s5_radius.py</code>	<code>s_experiment_rows.csv</code> <code>s5_radius_sweep.csv</code>

Figures are built by the matching `fig_*.py`. Experiment 2's probe log records per-probe coordinates, response, true label, fitted responsibility, trim mask and filter mask, so any downstream arm — oracle / soft / hard labels, trim on-off, filter on-off — is reconstructable as post-processing without a rerun. Experiment 3 requires the `fooling-lime-shap` repository as a sibling check-out, plus `xgboost`; Experiment 1 uses `dip-test` where available and a seeded Monte-Carlo null otherwise.

19 Open items these runs created

- Interval coverage on $(\hat{n}, \hat{c}, \hat{t})$ — never measured, and it is what turns a point estimate into a claim. [\[OPEN\]](#)
- Intrinsic-dimension scaling of orientation error. [\[OPEN\]](#)
- The no-information guard — required, not yet designed. [\[OPEN\]](#)
- The lattice account of the Communities & Crime miss — directionally supported, not established. [\[OPEN\]](#)
- A discrete-response Stage 3 variant, if the Slack datasets are to appear. [\[OPEN\]](#)